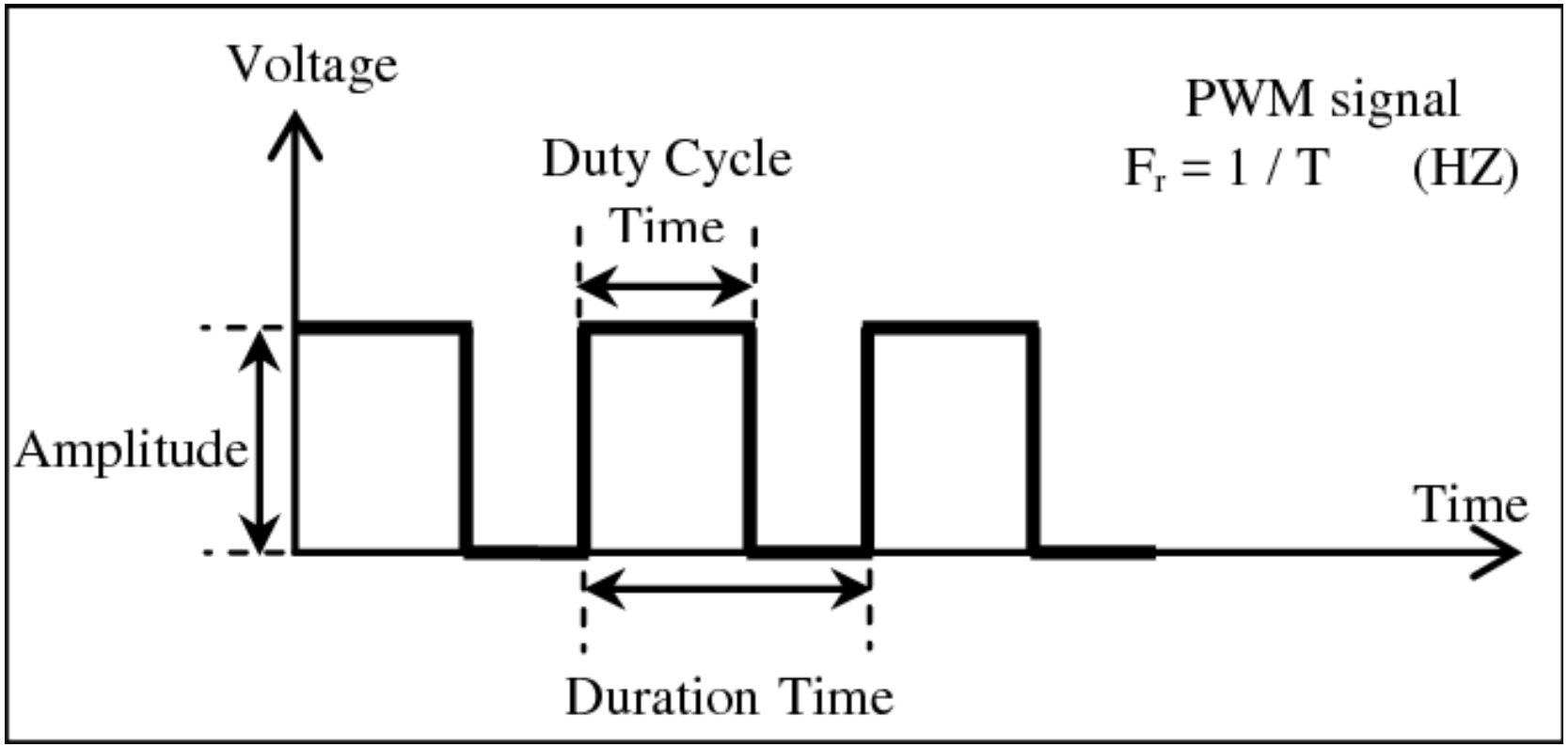
Università di Genova

Embedded Systems

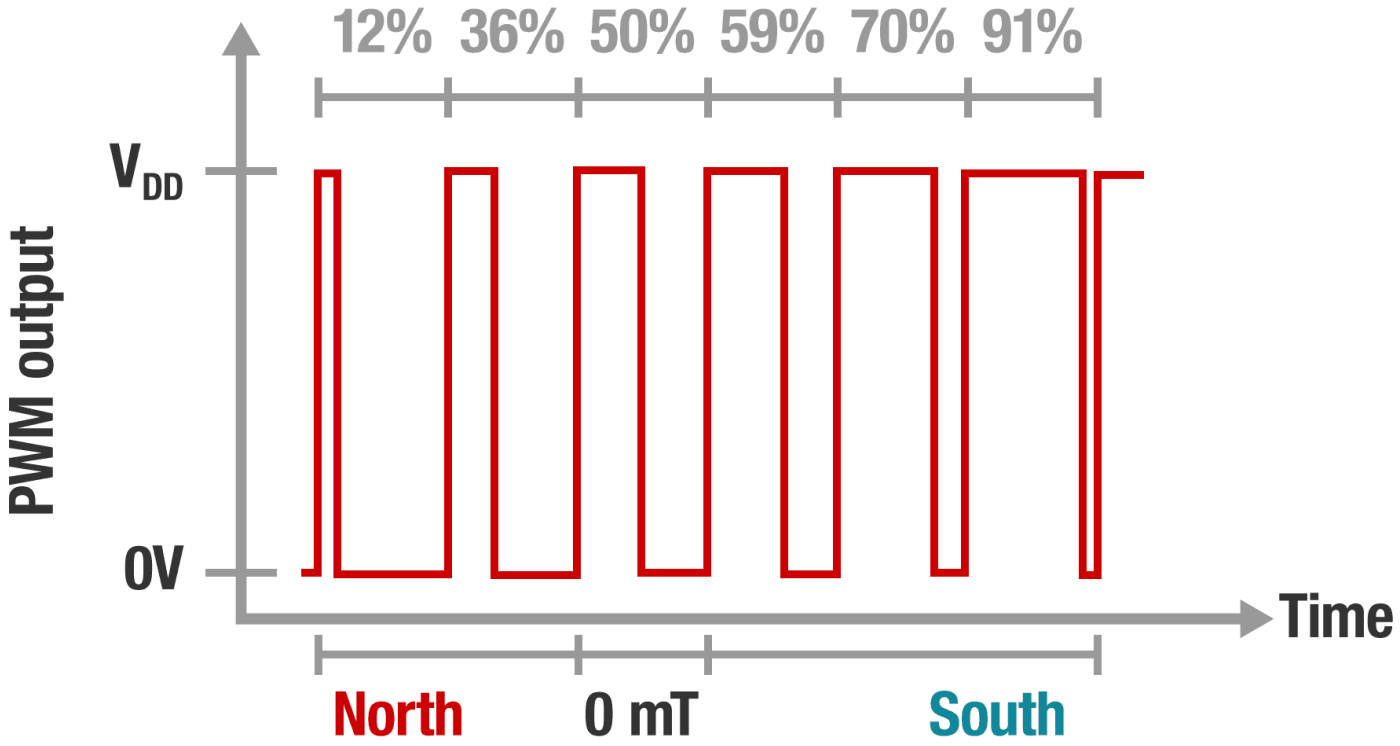
PWM and Output Compare

Prof. Enrico Simetti

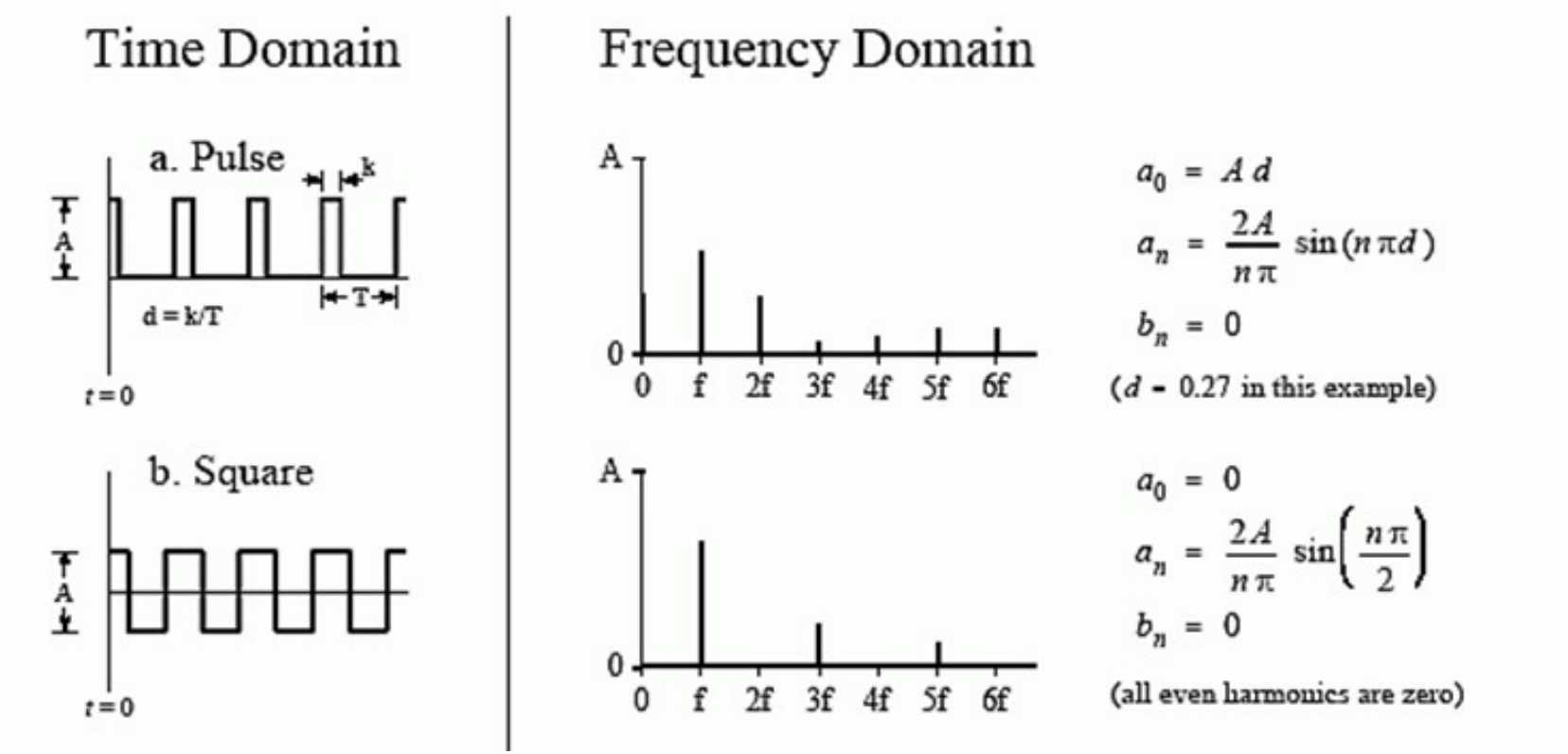
What is a Pulse Width Modulated Signal



PWM as an Encoding Method



PWM Frequency Spectrum



$a_0 = A\,d$

$a_n = \frac{2A}{n\pi}\sin(n\pi d)$

$b_n = 0$

(d = 0.27 in this example)

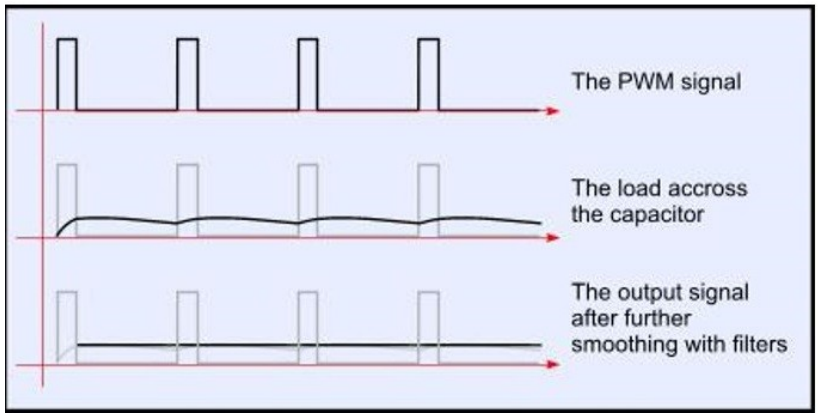
$a_0 = 0$

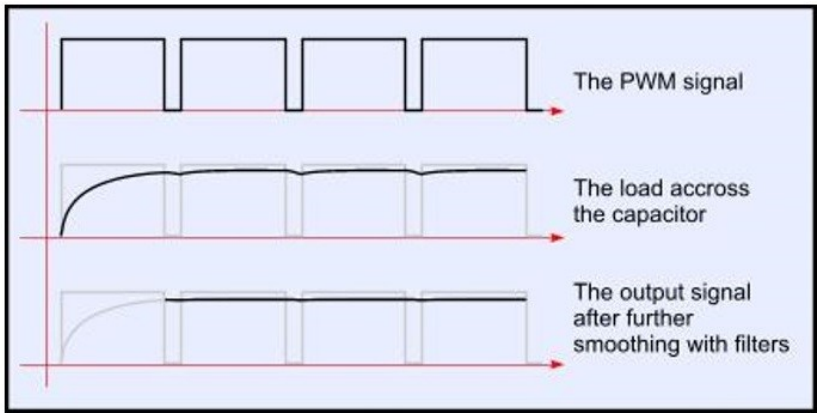
$a_n = \frac{2A}{n\pi}\sin\left(\frac{n\pi}{2}\right)$

$b_n = 0$

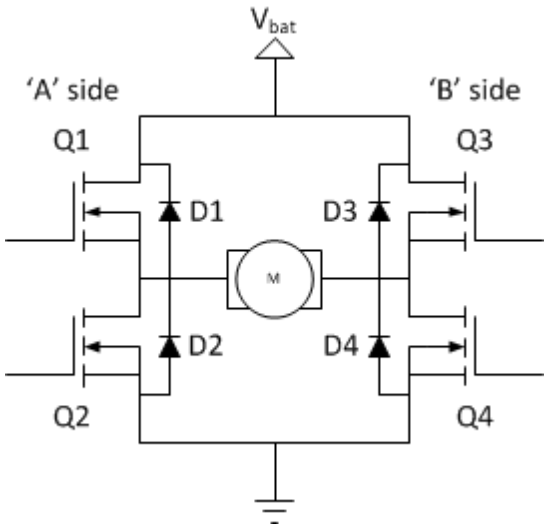
(all even harmonics are zero)

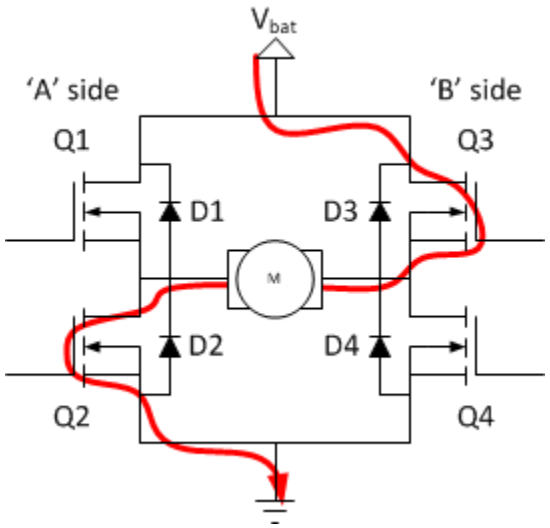
PWM with Low Pass Filter

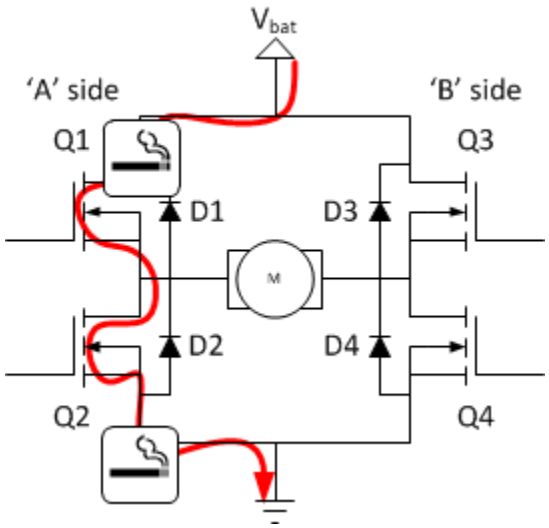




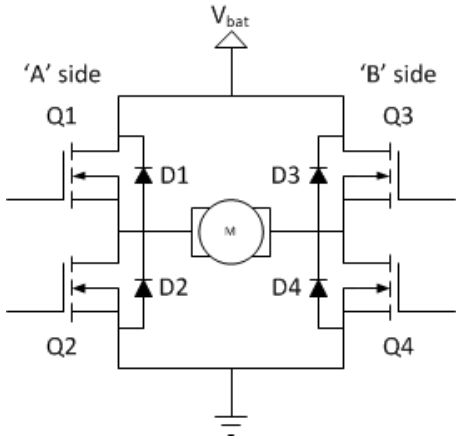
PWM and Motor Control

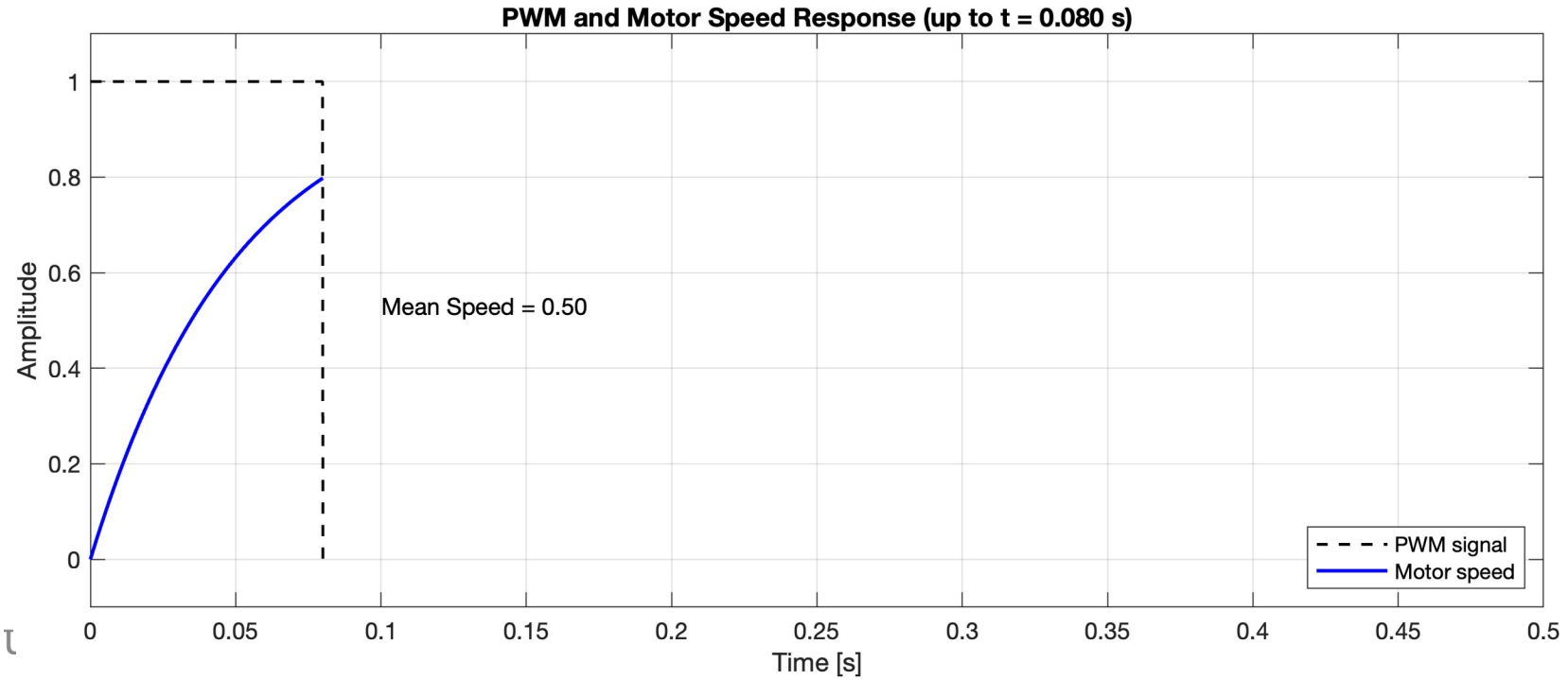




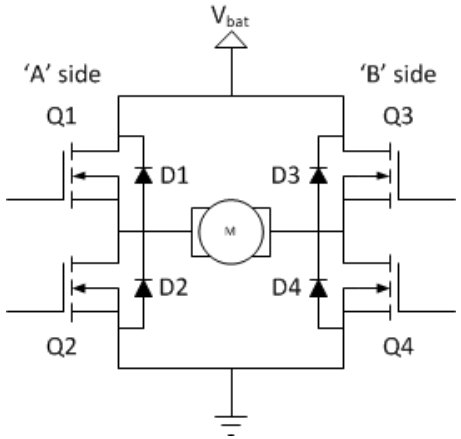


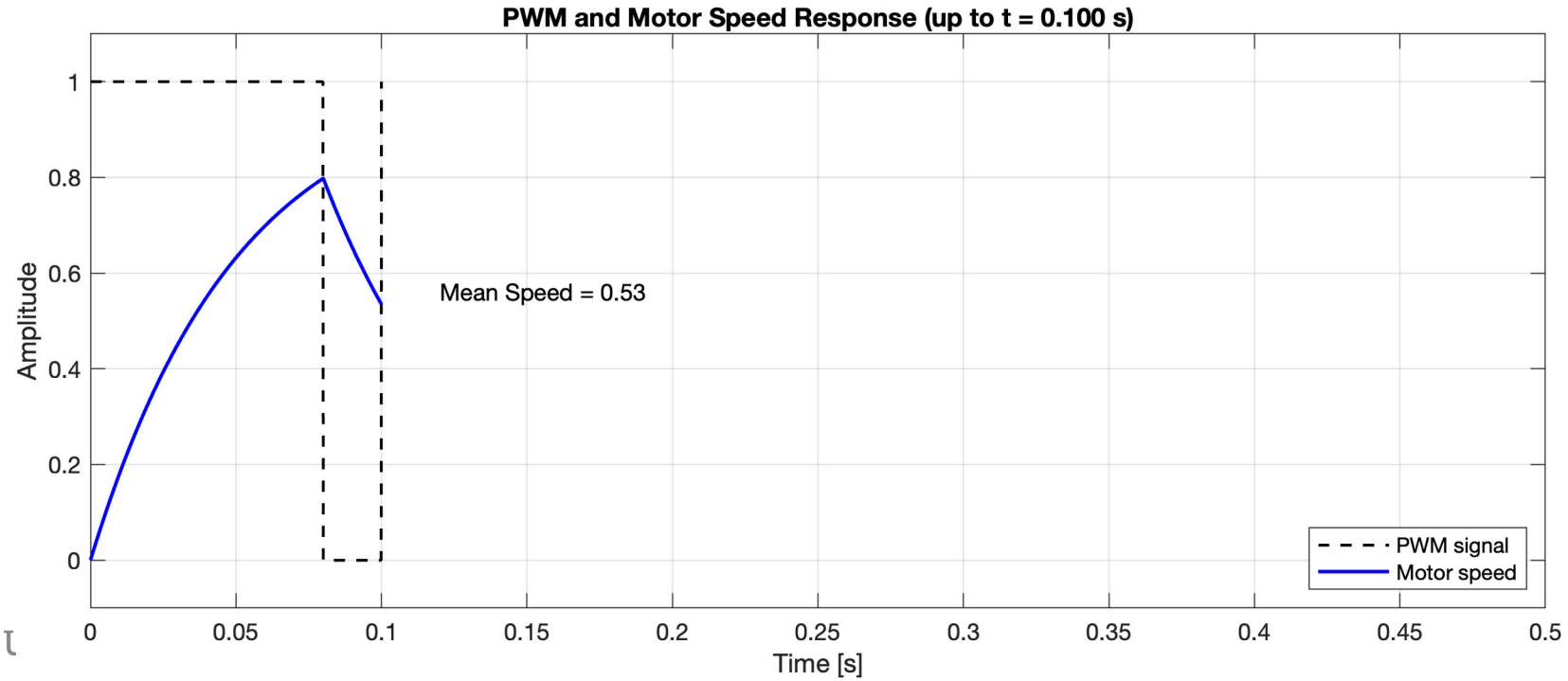
PWM and Motor Control



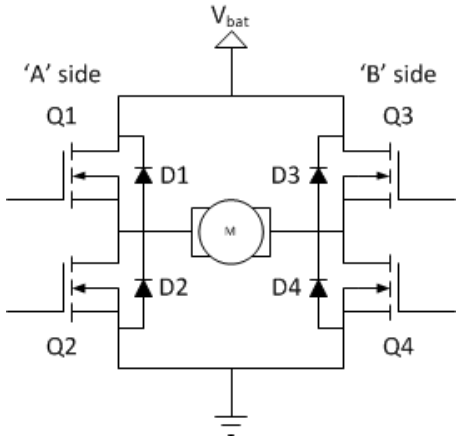


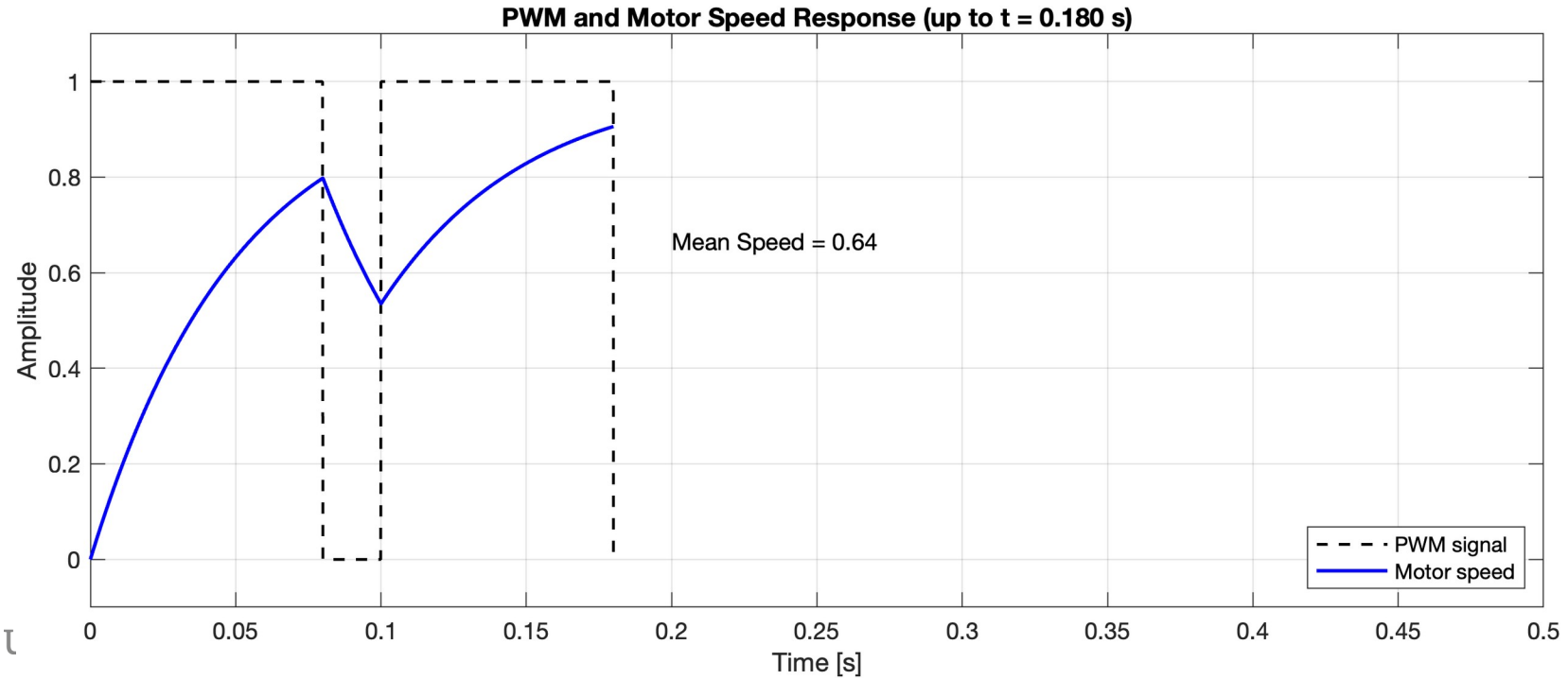
PWM and Motor Control



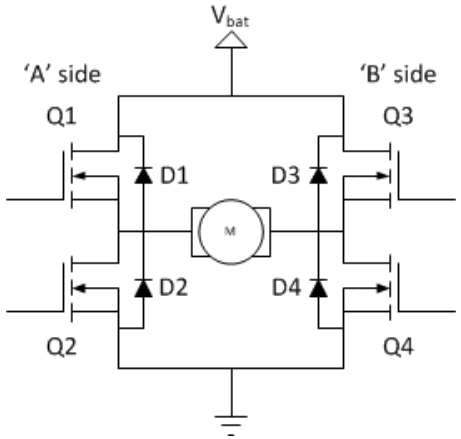


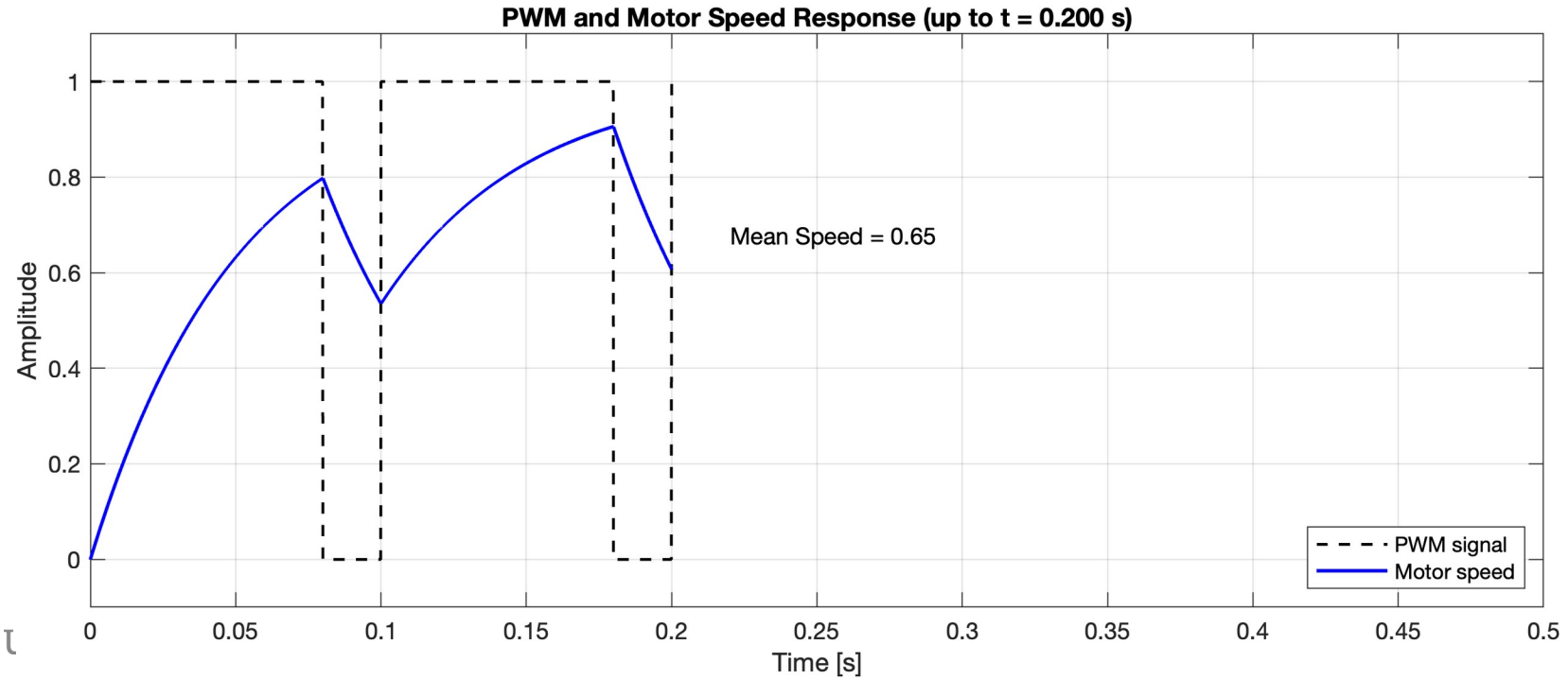
PWM and Motor Control



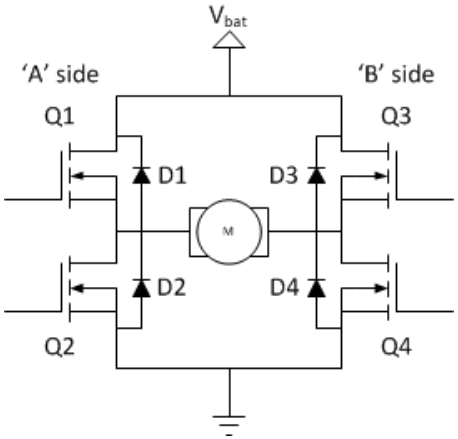


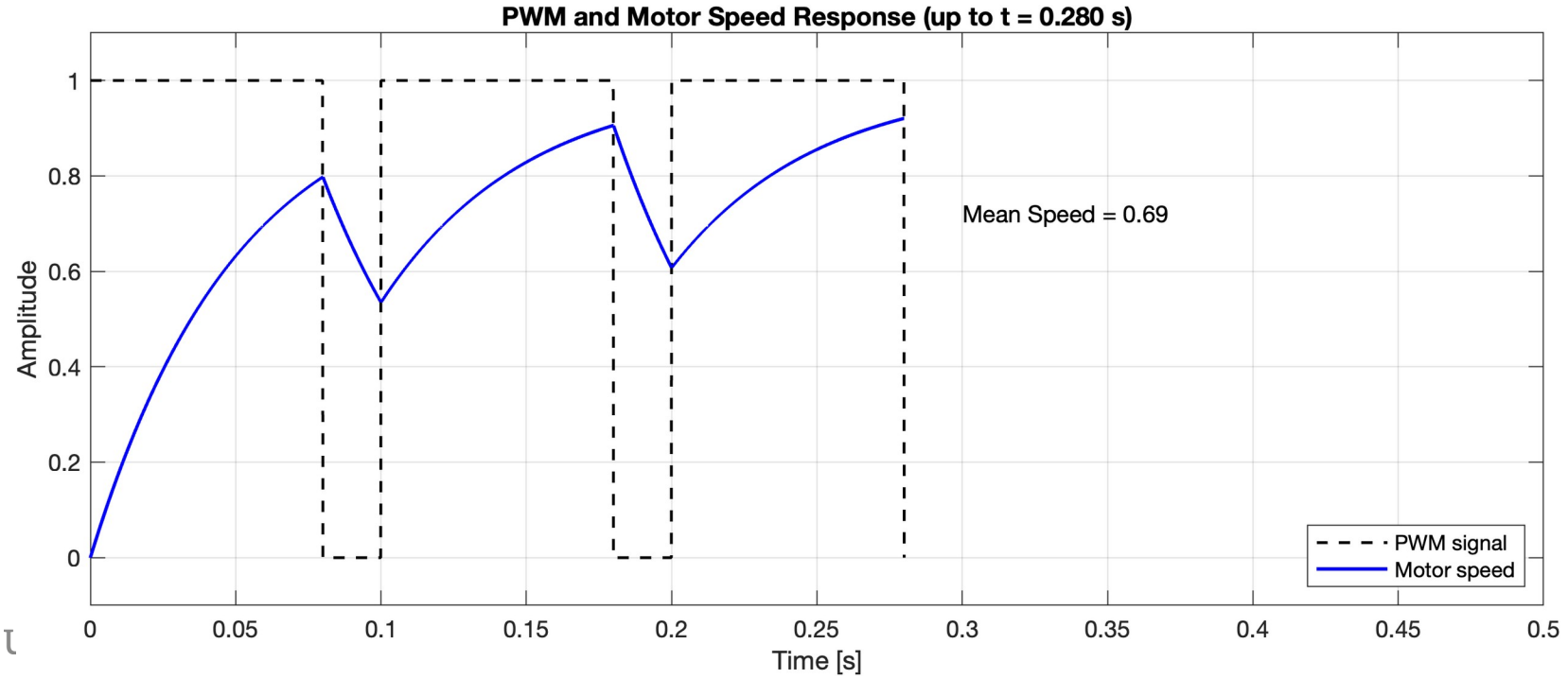
PWM and Motor Control



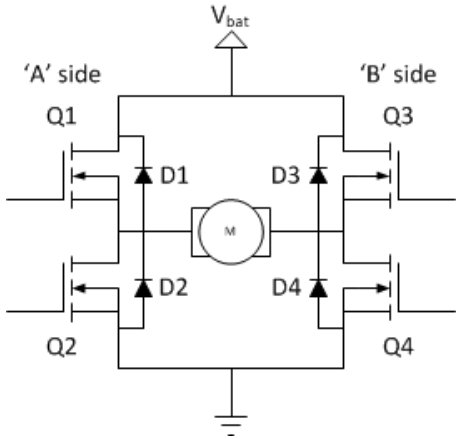


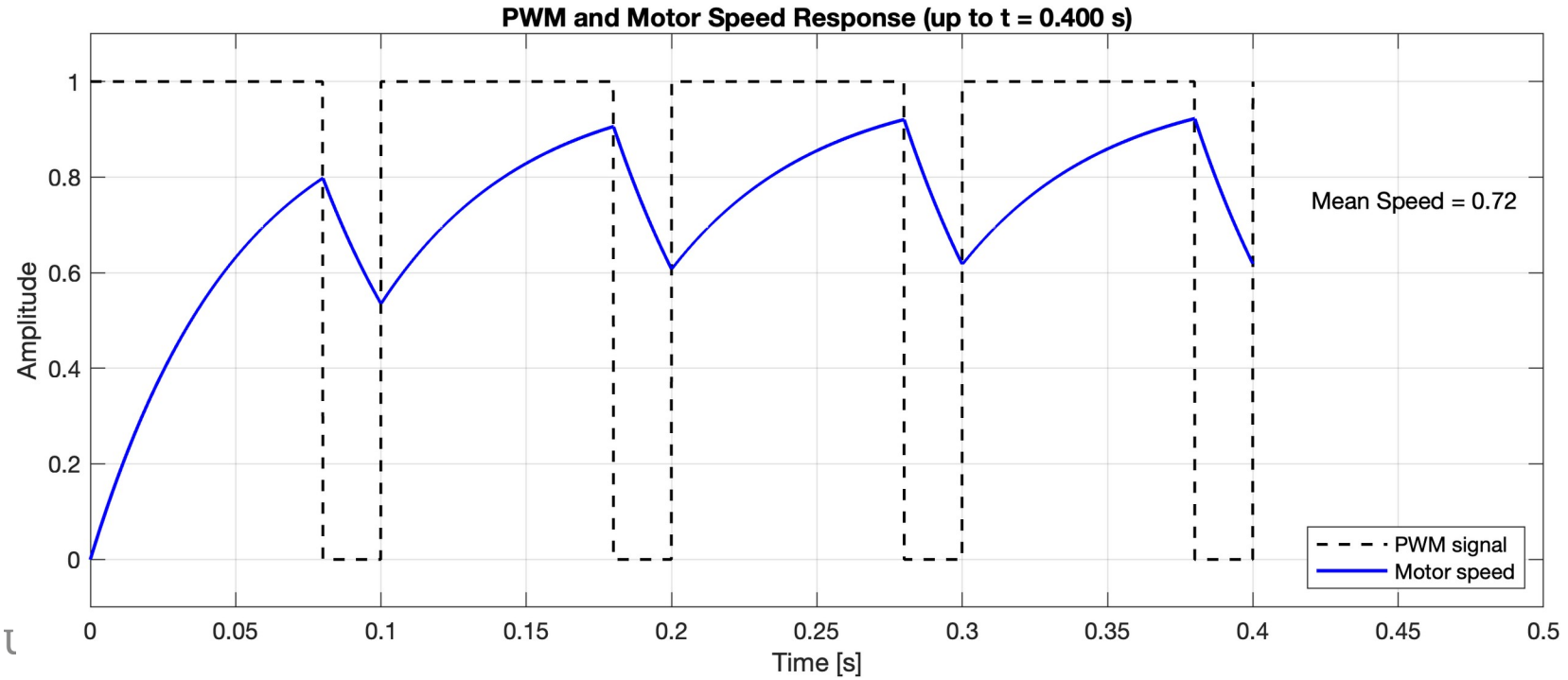
PWM and Motor Control



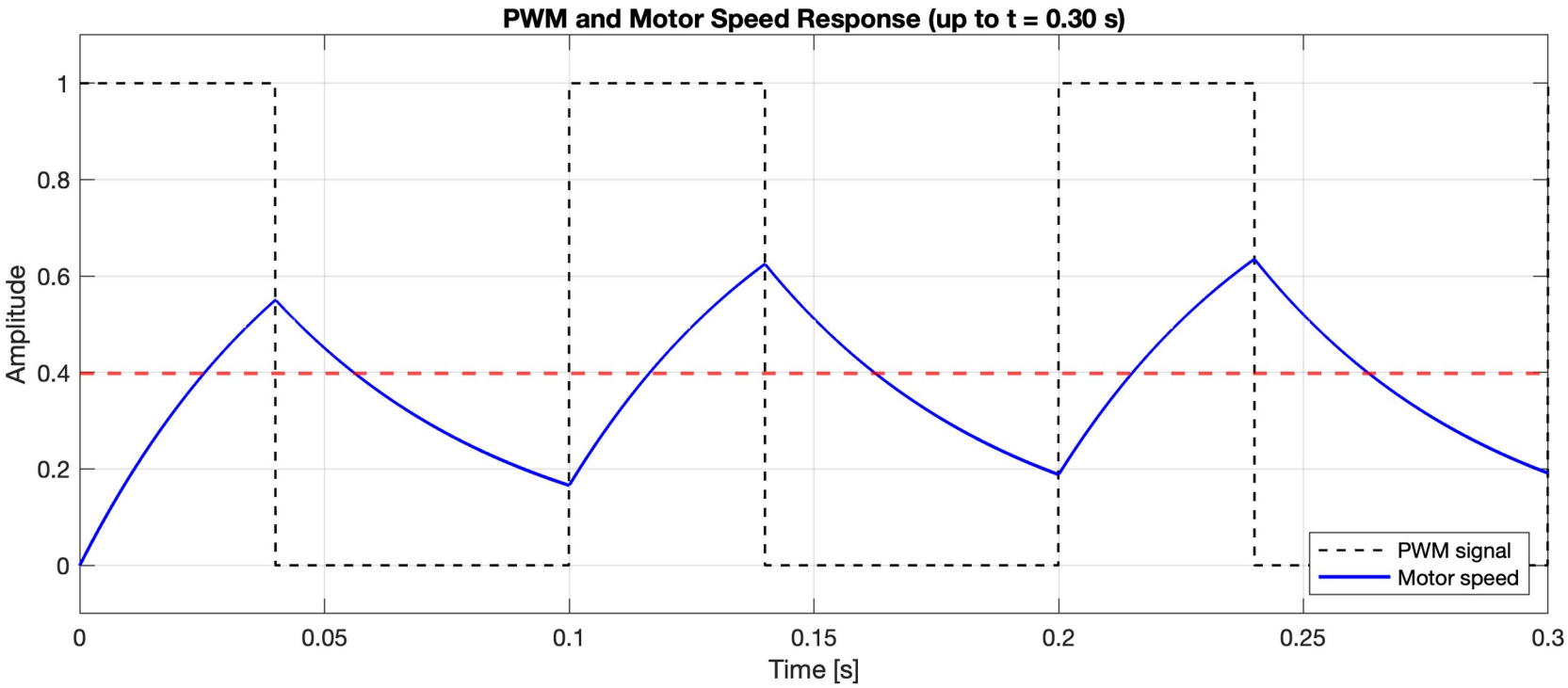


PWM and Motor Control

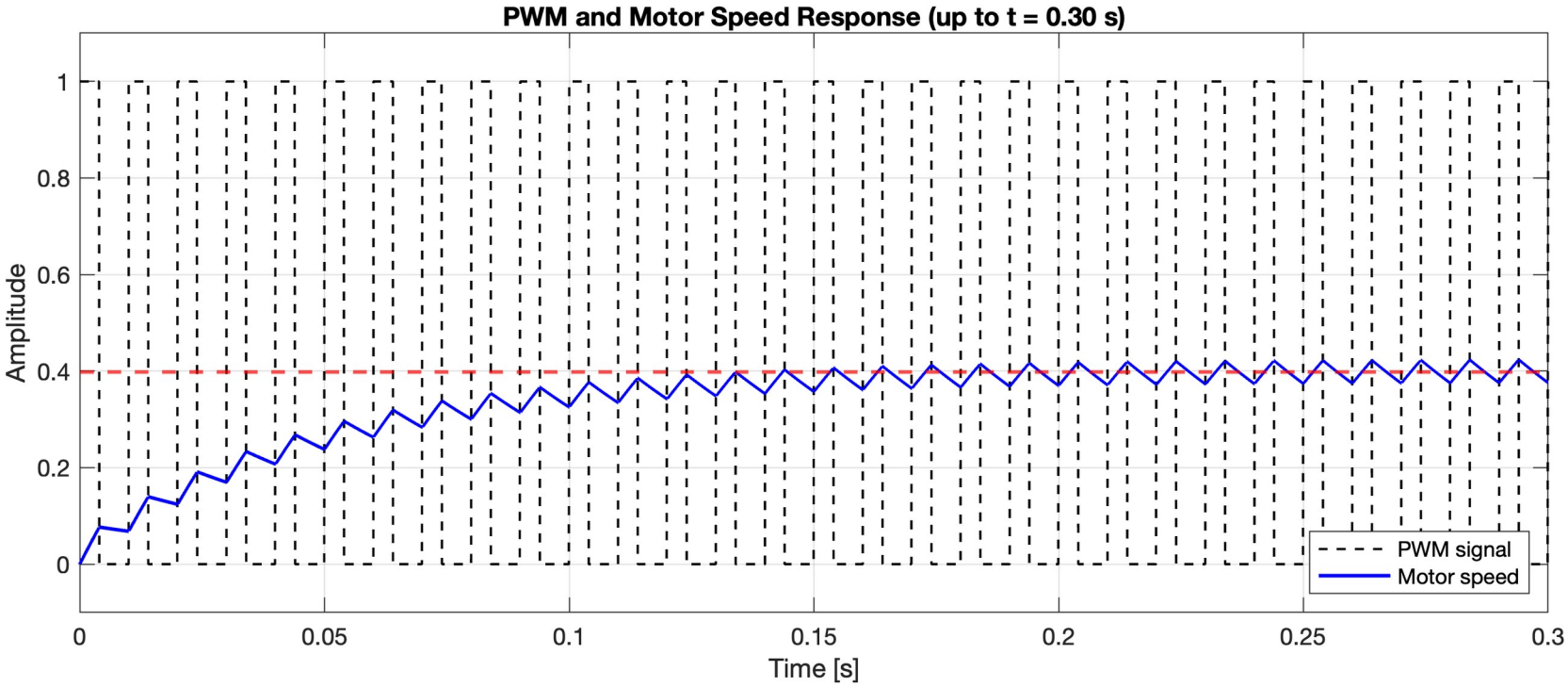




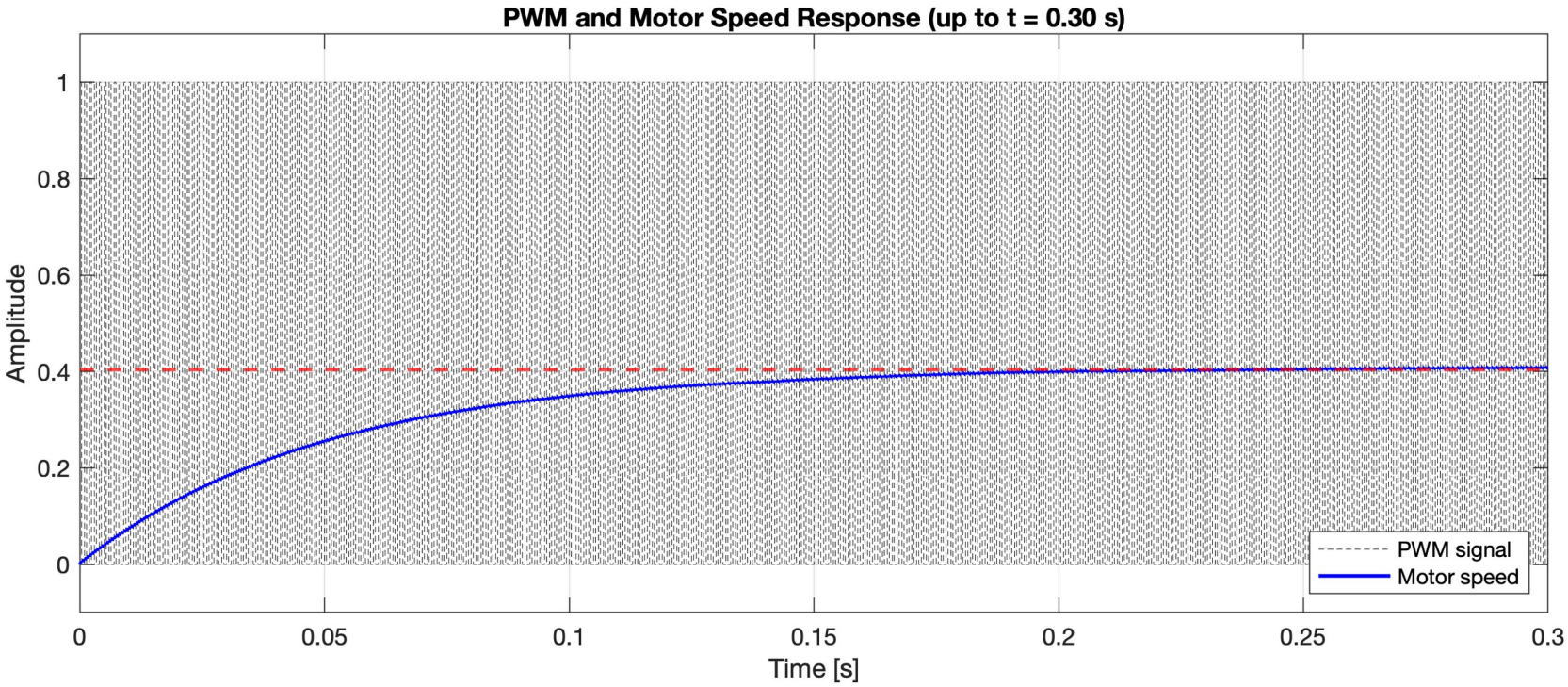
Example: PWM 40% DC and 10 Hz



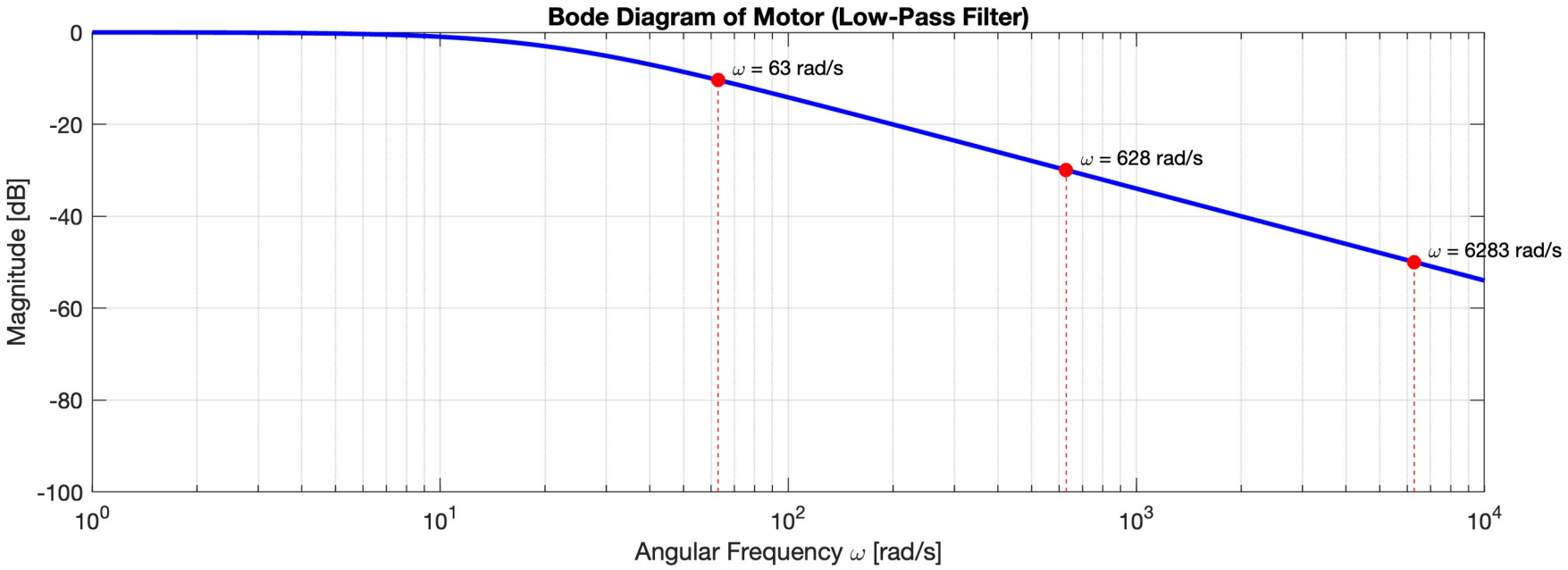
Example: PWM 40% DC and 100 Hz

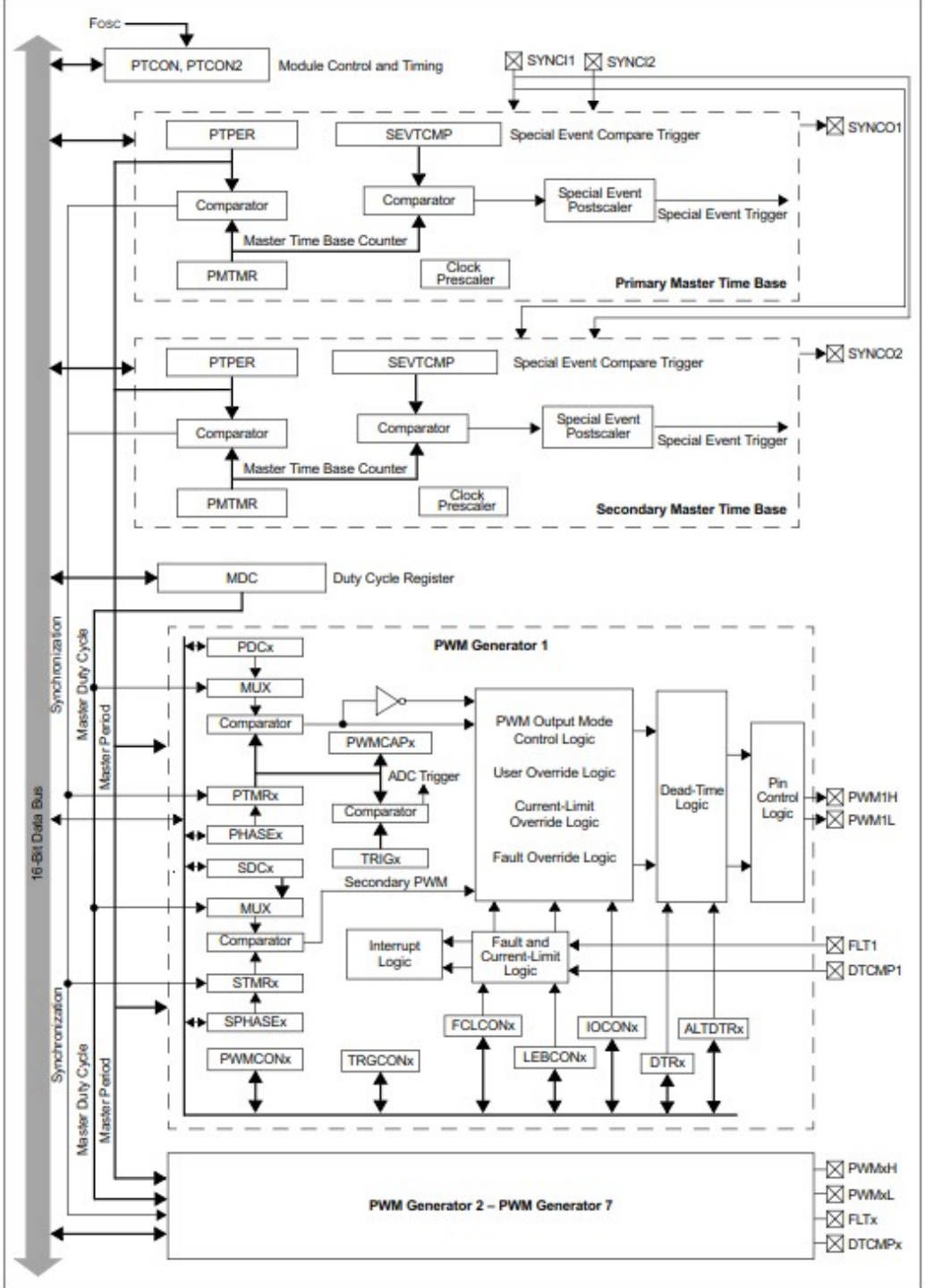


Example: PWM 40% DC and 1000 Hz

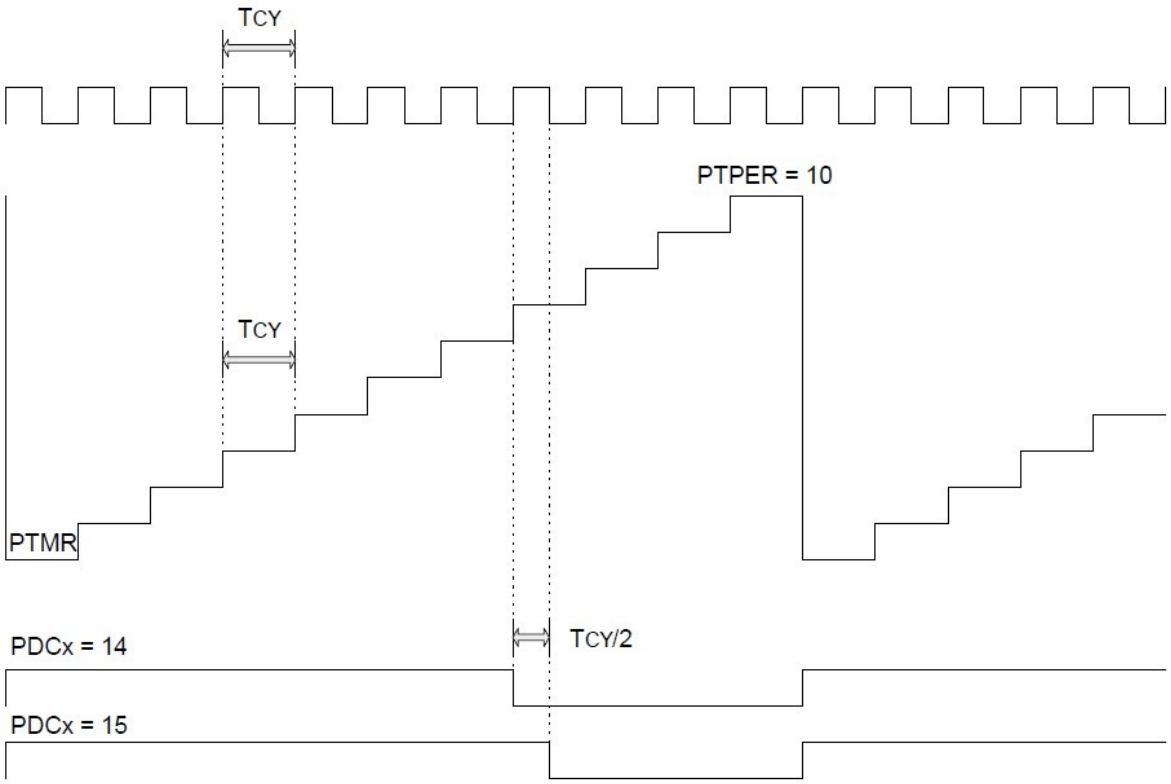


Bode Diagram




Fosc
PTCON, PTCON2
Module Control and Timing
SYNCI1
SYNCI2
PTPER
SEVTCMP
Special Event Compare Trigger
SYNCO1
Comparator
Comparator
Special Event Postscaler
Special Event Trigger
Master Time Base Counter
PMTMR
Clock Prescaler
Primary Master Time Base
PTPER
SEVTCMP
Special Event Compare Trigger
SYNCO2
Comparator
Comparator
Special Event Postscaler
Special Event Trigger
Master Time Base Counter
PMTMR
Clock Prescaler
Secondary Master Time Base
MDC
Duty Cycle Register
16-Bit Data Bus
Synchronization
Master Duty Cycle
Master Period
PWM Generator 1
PDCx
MUX
Comparator
PWMCAPx
ADC Trigger
PTMRx
Comparator
PHASEx
TRIGx
SDCx
Secondary PWM
MUX
Comparator
STMRx
SPHASEx
PWMCONx
TRGCONx
PWM Output Mode Control Logic
User Override Logic
Current-Limit Override Logic
Fault Override Logic
Dead-Time Logic
Pin Control Logic
PWM1H
PWM1L
Interrupt Logic
Fault and Current-Limit Logic
FLT1
DTCMP1
FCLCONx
LEBCONx
IOCONx
DTRx
ALTDTRx
PWM Generator 2 – PWM Generator 7
PWMxH
PWMxL
FLTx
DTCMPx

PWM DC Generation with PDC



PTCON Register

Register 3-1: PTCON: PWMx Time Base Control Register

R/W-0	U-0	R/W-0	HS/HC-0	R/W-0	R/W-0	R/W-0	R/W-0
PTEN(3)	—	PTSIDL	SESTAT	SEIEN	EIPU(1)	SYNCPOL(1,2)	SYNCOEN(1,2)
bit 15							bit 8

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
SYNCEN(1,2)	SYNCSRC<2:0>(1,2)			SEVTPS<3:0>(1)			
bit 7							bit 0

PTCON: PWM time base Control register.

- PTEN: turns on the PWM time base module

PTCON2 Register

PCLKDIV<2:0>:

111 = Reserved
110 = Divide-by-64
101 = Divide-by-32
100 = Divide-by-16
011 = Divide-by-8
010 = Divide-by-4
001 = Divide-by-2
000 = Divide-by-1

REGISTER 16-2: PTCON2: PWM PRIMARY MASTER CLOCK DIVIDER SELECT REGISTER 2

U-0	U-0	U-0	U-0	U-0	U-0	U-0	U-0
—	—	—	—	—	—	—	—
bit 15							bit 8

U-0	U-0	U-0	U-0	U-0	R/W-0	R/W-0	R/W-0
—	—	—	—	—	PCLKDIV<2:0>(1)		
bit 7							bit 0

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'	
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared	x = Bit is unknown

PTCON2: PWM primary master clock divider selector register.

- PCLKDIV: choose PWM timer prescaler

IOCONx Register

REGISTER 16-19: IOCONx: PWMx I/O CONTROL REGISTER

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
PENH	PENL	POLH	POLL	PMOD<1:0>(1)		OVRENH	OVRENL
bit 15							bit 8

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
OVRDAT<1:0>		FLTDAT<1:0>		CLDAT<1:0>		SWAP	OSYNC
bit 7							bit 0

IOCONx: PWM Pins Control register

- PENH: selects if the pin should be used as PWM output
- PENL: selects if the pin should be used as PWM output

Always check the correct pins on the dspic33EP512MU810 layout

PDCx and PTPER Registers

$$PTPER = \frac{F_{CY}}{F_{PWM} \bullet (\text{PTMR Prescaler})} - 1$$

Example:

$F_{CY} = 20$ MHz

$F_{PWM} = 20{,}000$ Hz

PTMR Prescaler = 1:1

$$PTPER = \frac{20{,}000{,}000}{20{,}000 \bullet 1} - 1$$

$$= 1000 - 1$$

$$= 999$$

PDCx: used to specify the duty cycle

PTPER: used to specify the time period **(15 bits register!)**

Output Compare

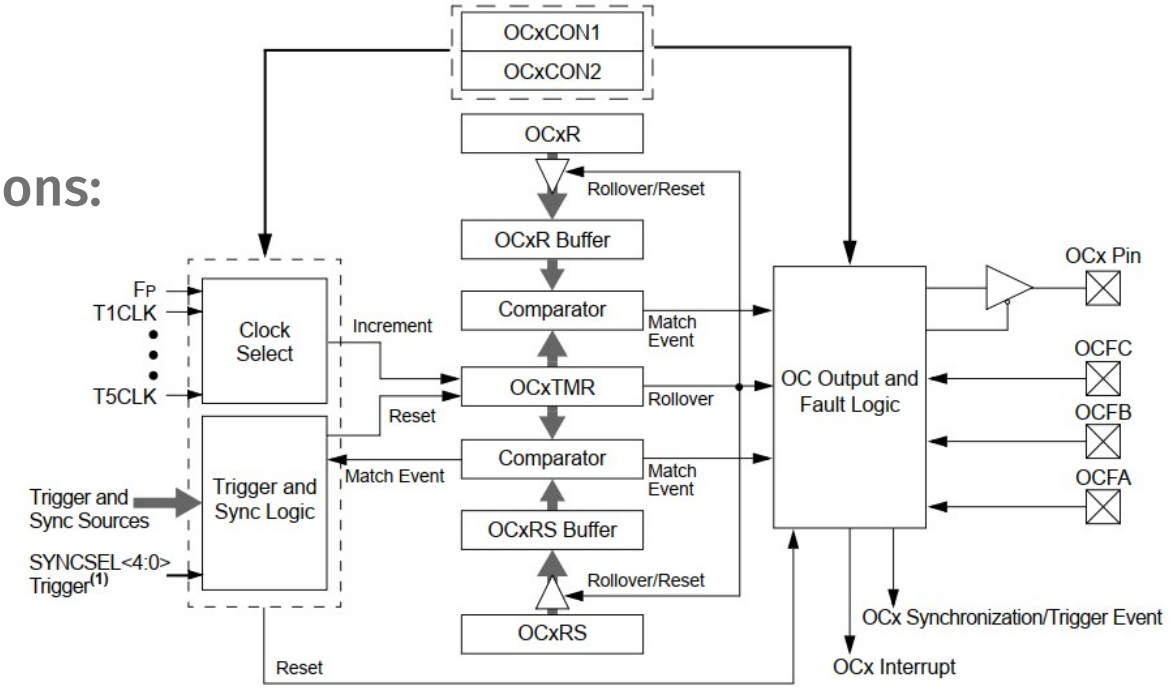
The Output Compare module can compare the value of a selected time base with the value of one or two compare registers (depending on the Operation mode selected).

Furthermore, it can generate a single output pulse, or a train of output pulses, on a compare match event. Like most dsPIC peripherals, it also can generate interrupts on-compare match events.

Output Compare

3 Modes of operations:

- Single Compare
- Dual Compare
- PWM



OCxCON1 Register

REGISTER 15-1: OCxCON1: OUTPUT COMPARE x CONTROL REGISTER 1

U-0	U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
—	—	OCSIDL	OCTSEL<2:0>			ENFLTC	ENFLTB
bit 15							bit 8

R/W-0	R/W-0, HSC	R/W-0, HSC	R/W-0, HSC	R/W-0	R/W-0	R/W-0	R/W-0
ENFLTA	OCFLTC	OCFLTB	OCFLTA	TRIGMODE	OCM<2:0>		
bit 7							bit 0

OCxCON1: Control Register 1

- OCTSEL: selects the input clock source for the OCx module
- OCM: selects the OC modality for the OCx module

OCxCON1 Register

OCTSEL<2:0>: Output Compare x Clock Select bits

111 = Peripheral clock (FP)
110 = Reserved
101 = Reserved
100 = Clock source of T1CLK is the clock source of OCx (only the synchronous clock is supported)
011 = Clock source of T5CLK is the clock source of OCx
010 = Clock source of T4CLK is the clock source of OCx
001 = Clock source of T3CLK is the clock source of OCx
000 = Clock source of T2CLK is the clock source of OCx

OCM<2:0>: Output Compare Mode Select bits

111 = Center-Aligned PWM mode: Output set high when OCxTMR = OCxR and set low when OCxTMR = OCxRS(1)
110 = Edge-Aligned PWM mode: Output set high when OCxTMR = 0 and set low when OCxTMR = OCxR(1)
101 = Double Compare Continuous Pulse mode: Initializes OCx pin low, toggles OCx state continuously on alternate matches of OCxR and OCxRS
100 = Double Compare Single-Shot mode: Initializes OCx pin low, toggles OCx state on matches of OCxR and OCxRS for one cycle
011 = Single Compare mode: Compares events with OCxR, continuously toggles OCx pin
010 = Single Compare Single-Shot mode: Initializes OCx pin high, compares event with OCxR, forces OCx pin low
001 = Single Compare Single-Shot mode: Initializes OCx pin low, compares event with OCxR, forces OCx pin high
000 = Output compare channel is disabled

Suggested setting is to use the peripheral clock as source. If you want to choose an internal timer, you need to configure it properly. Suggested setting for OCM is Edge-Aligned.

OCxCON2 Register

REGISTER 15-2: OCxCON2: OUTPUT COMPARE x CONTROL REGISTER 2

R/W-0	R/W-0	R/W-0	R/W-0	U-0	U-0	U-0	R/W-0
FLTMD	FLTOUT	FLTTRIEN	OCINV	—	—	—	OC32
bit 15							bit 8

R/W-0	R/W-0 HS	R/W-0	R/W-0	R/W-1	R/W-1	R/W-0	R/W-0
OCTRIG	TRIGSTAT	OCTRIS	SYNCSEL<4:0>				
bit 7							bit 0

OCxCON2: Control Register 2

- SYNCSEL: Selects synchronization source for the OCx module

The sync source choice should be coherent with the clock source. If peripheral clock is used, then no sync source is required

OCxCON2 Register

SYNCSEL<4:0>: Trigger/Synchronization Source Selection bits

11111 = No Sync or Trigger source for OCx
11110 = INT2 pin synchronizes or triggers OCx
11101 = INT1 pin synchronizes or triggers OCx
11100 = Reserved
11011 = ADC1 module synchronizes or triggers OCx
11010 = CMP3 module synchronizes or triggers OCx
11001 = CMP2 module synchronizes or triggers OCx
11000 = CMP1 module synchronizes or triggers OCx
10111 = IC8 module synchronizes or triggers OCx
10110 = IC7 module synchronizes or triggers OCx
10101 = IC6 module synchronizes or triggers OCx
10100 = IC5 module synchronizes or triggers OCx
10011 = IC4 module synchronizes or triggers OCx
10010 = IC3 module synchronizes or triggers OCx
10001 = IC2 module synchronizes or triggers OCx
10000 = IC1 module synchronizes or triggers OCx
01111 = Timer5 synchronizes or triggers OCx
01110 = Timer4 synchronizes or triggers OCx
01101 = Timer3 synchronizes or triggers OCx
01100 = Timer2 synchronizes or triggers OCx (default)
01011 = Timer1 synchronizes or triggers OCx
01010 = No Sync or Trigger source for OCx
01001 = OC9 module synchronizes or triggers OCx(1,2)
01000 = OC8 module synchronizes or triggers OCx(1,2)
00111 = OC7 module synchronizes or triggers OCx(1,2)
00110 = OC6 module synchronizes or triggers OCx(1,2)
00101 = OC5 module synchronizes or triggers OCx(1,2)
00100 = OC4 module synchronizes or triggers OCx(1,2)
00011 = OC3 module synchronizes or triggers OCx(1,2)
00010 = OC2 module synchronizes or triggers OCx(1,2)
00001 = OC1 module synchronizes or triggers OCx(1,2)
00000 = No Sync or Trigger source for OCx

OCxR and OCxRS Registers

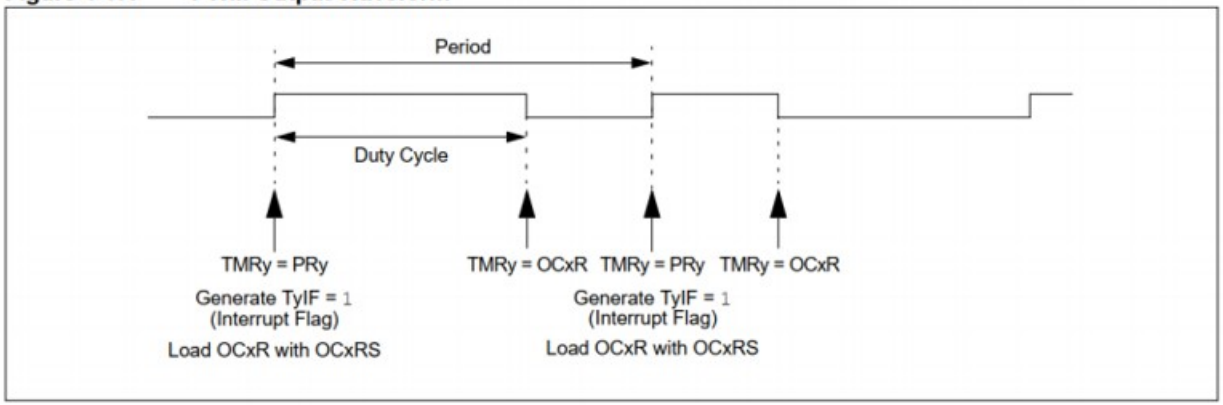
OCxR: primary OCx module register

OCxRS: secondary OCx module register

These registers are essential for the duty cycle configuration

PWM Generation with OC (Timer Source)

Figure 4-17: PWM Output Waveform



$$\text{PWM Period} = [(\text{PRy}) + 1] \bullet T_{CY} \bullet (\text{TMRy Prescale Value})$$

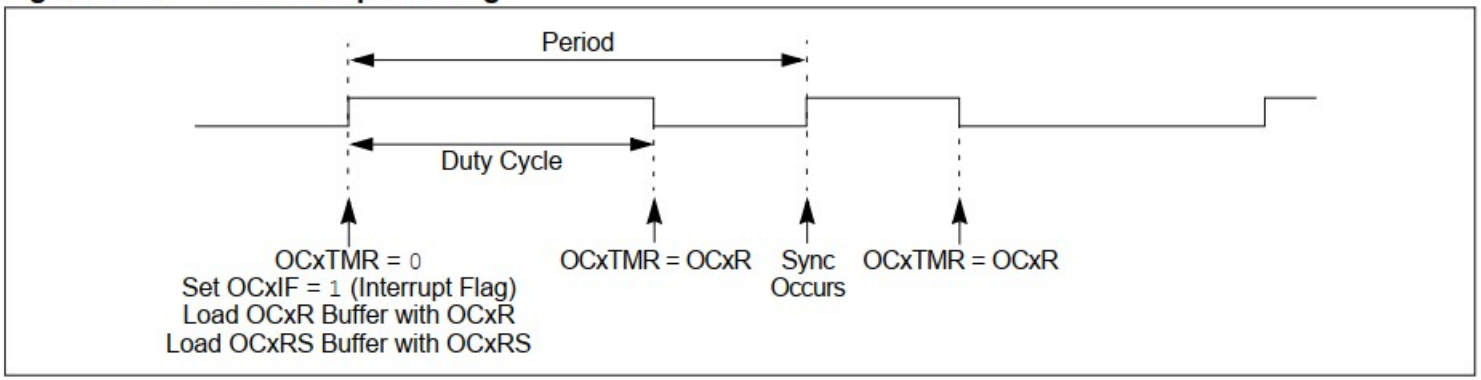
$$\text{PWM Frequency} = 1/[\text{PWM Period}]$$

Important! This formula is only valid if the clock and sync sources come from a timer Ty.

PWM Generation with OC (Internal Source)

Suggested setting: Configure the OCx module clock source as the system clock, so that $T_{PWM} = T_{cy}$

Figure 3-13: PWM Output Timing

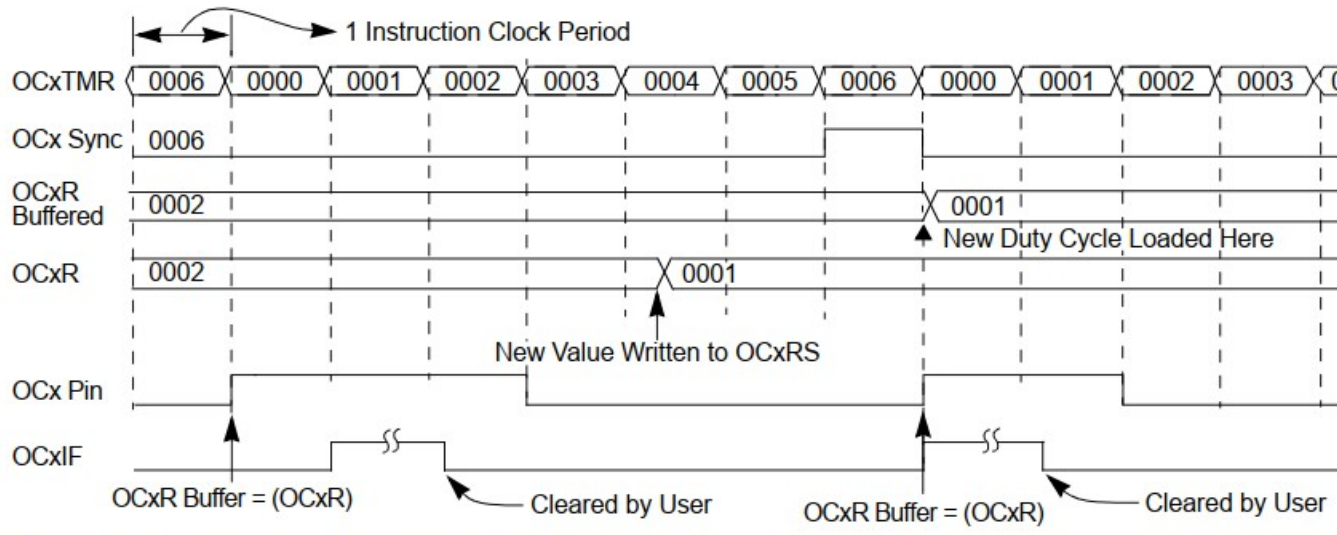


When the SYNCSEL bits (OCxCON2) are configured as 0b11111 (0x1F), they make the internal timer reset when it reaches the value of OCxRS, making the OCx module use its own Sync signal.

This example refers to Edge-Aligned PWM mode.

PWM Generation with OC (Internal Source)

Figure 3-14: PWM Output Timing



Note: In this example, one instruction clock period comprises two Fosc cycles.

```
OCxCON1bits.OCTSEL = 7;       // Internal clock
OCxCON2bits.SYNCSEL = 0x1F;   // No sync source
```

Buggy Motor Control

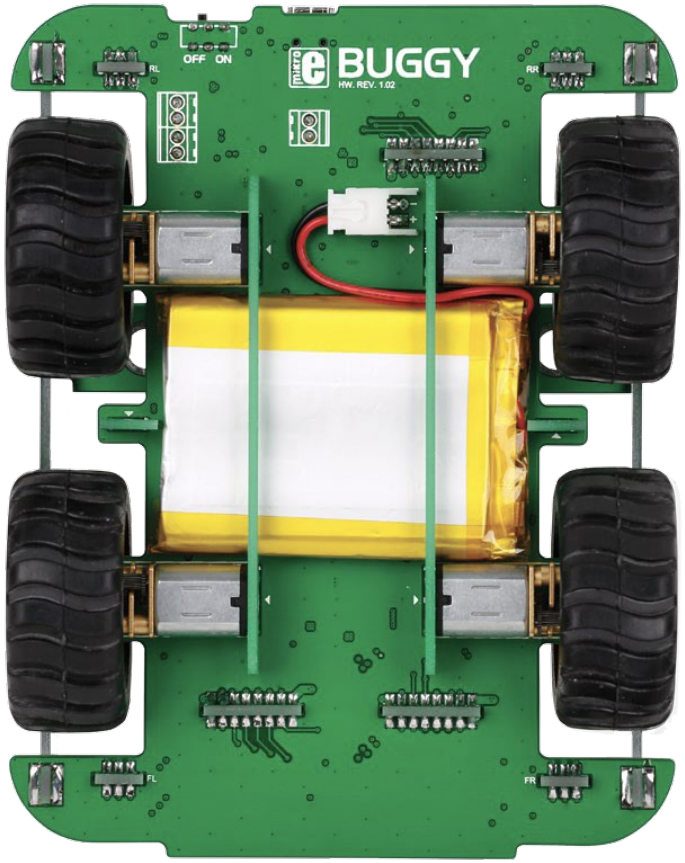
The Buggy has a differential motor drive. The four DC motors

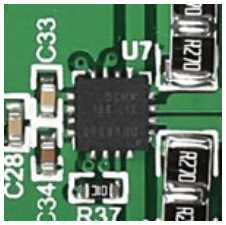
are split by left and right axis (controlled by DRV833RTY motor

drivers, U6 and U7, one for each side). Steering takes place when

you vary the relative rate of rotation between the left and right

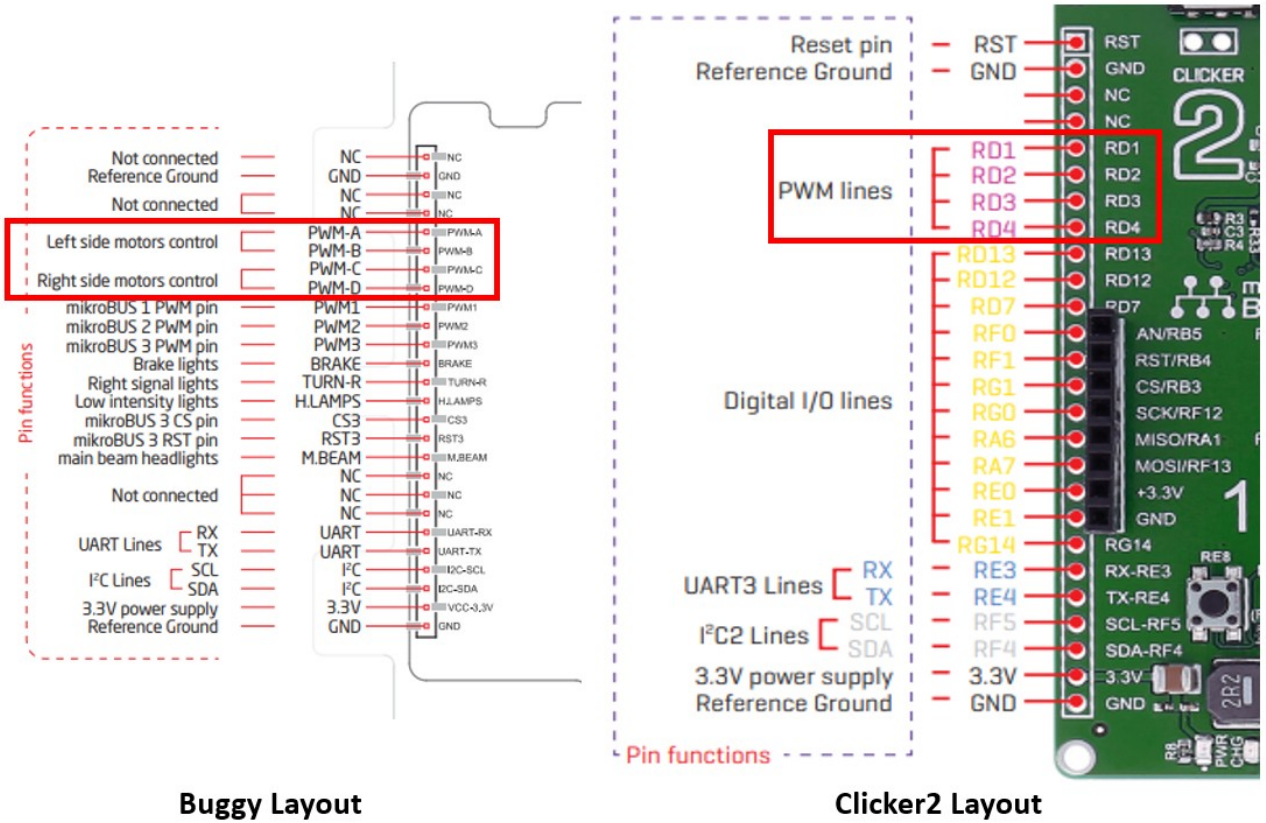
side.







Buggy Motor Control



Buggy Layout **Clicker2 Layout**

Remember to correctly reprogram the output pins on the dspic33 to match the correct PWM layout!

Buggy Motor Control

The output on the four PWM pins should be configured to match the following table (from the DRV8833 motor driver docs)

Table 2. PWM Control of Motor Speed

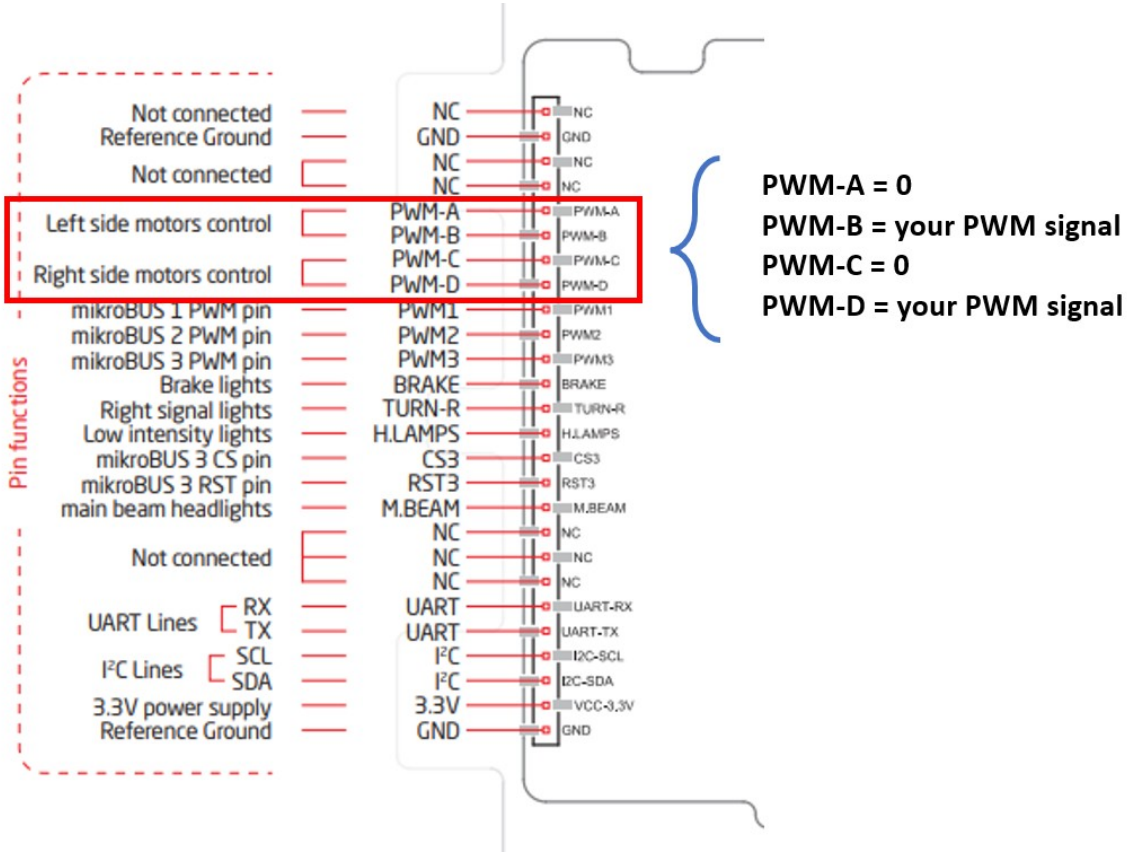
xIN1	xIN2	FUNCTION
PWM	0	Forward PWM, fast decay
0	PWM	Reverse PWM, fast decay

RD2 – RD4 **RD1 – RD3**

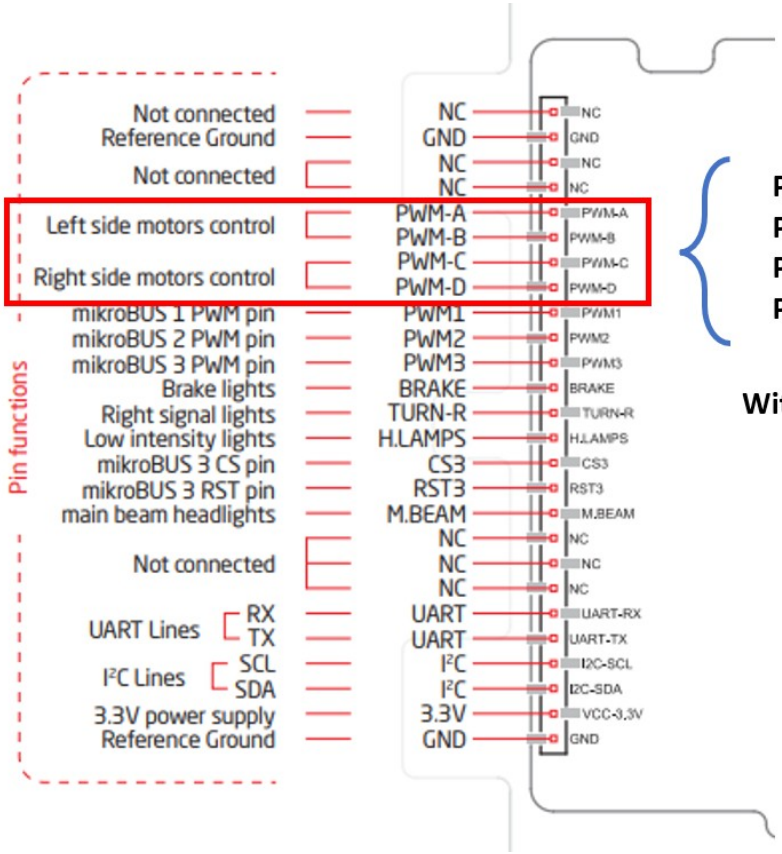
The duty cycle will modulate the speed of the two motors on each side, while the forward/backward rotation is specified on which pin is receiving the PWM signal.

Always connect the battery to provide the right amount of current to the motors! They will barely move if powered only via USB cable

Example of Forward Motion



Example of Right Turn while Moving Forward



PWM-A = 0
PWM-B = your PWM signal
PWM-C = 0
PWM-D = your PWM signal

With duty cycle B > duty cycle D

Assignment

- When button E8 is pressed, start generating a PWM signal with the Output Compare peripheral at a 10 kHz frequency to make the buggy move forward. When the same button E8 is pressed again, stop generating the PWM signal. Hint: the duty cycle range to enable motion is between 40% and 100%.
- When button E8 is pressed, start generating a PWM signal with the Output Compare peripheral at a 10 kHz frequency to make the buggy move clockwise in the forward direction. When the same button E8 is pressed again, stop generating the PWM signal.
- Control the buggy with two signals, speed and yaw rate that go from -100% to 100%. Implement a function that takes as input speed and yaw rate, and generates the two PWM for left and right. Test the function with different values for speed and jog when button E8 is pressed as before.



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